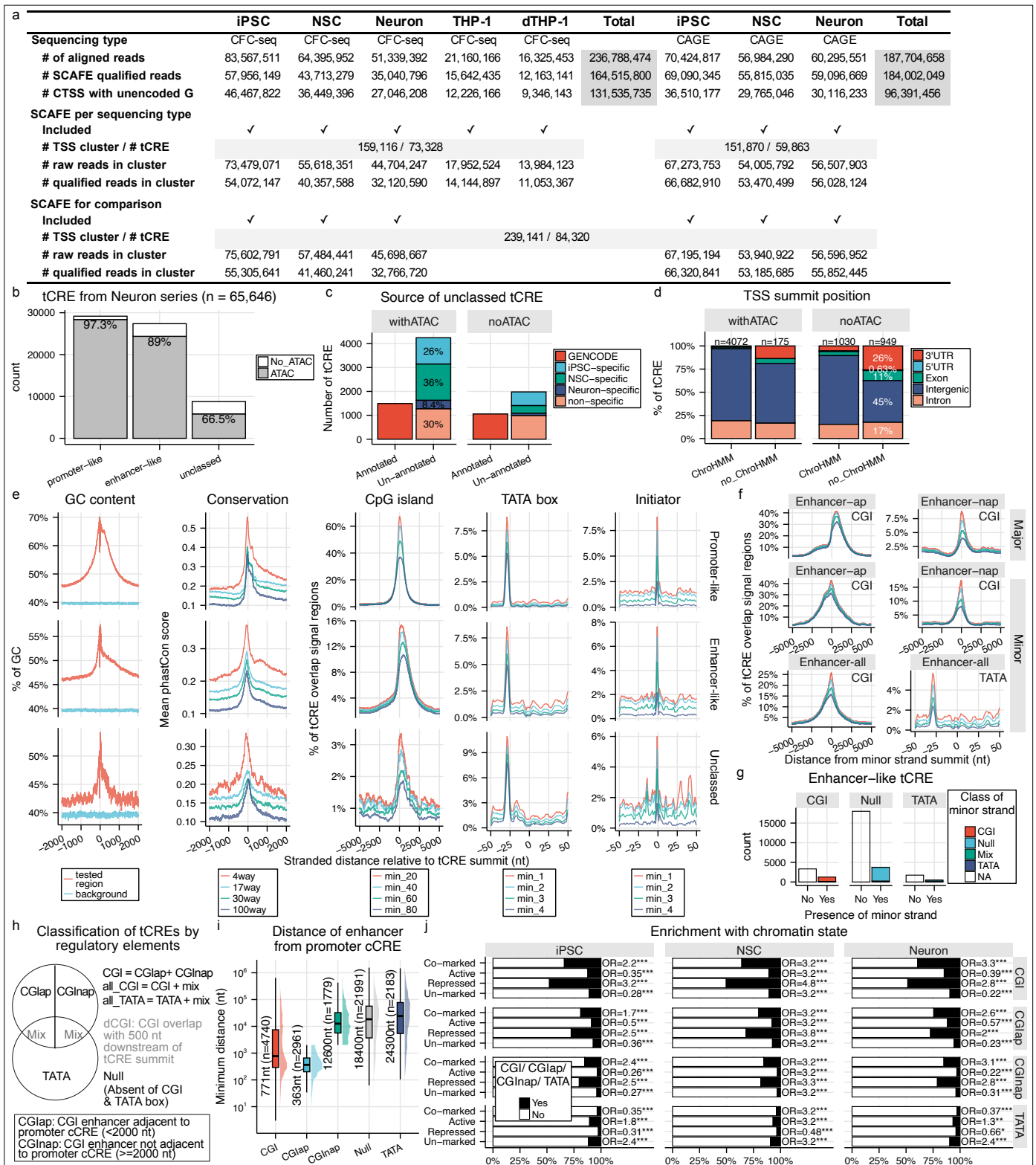




Extended Data Figure 2 | Identification of genuine TSSs, clusters, and tCREs from 5' ends of CFC-seq reads.

a, Summary of reads processed with SCAFE for identification of genuine TSS clusters and tCREs. **b**, Neuron-series tCREs overlap with ATAC-seq-defined open chromatin from the same samples. **c**, Unclassed tCREs were stratified by ACTA-seq support, GENCODE annotation status, and the originating cell type. **d**, Overlap of tCREs with chromatin states defined by ChromHMM using CUT&Tag data (H3K27ac, H3K27me3, H3K4me1, H3K4me3, CTCF) from three Neuron-series cell types. Colors indicate summit positions relative to GENCODE models. **e**, Genomic features of major-strand tCREs (n = 57,597) categorized as promoter-like, enhancer-like, or unclassified. Analyses include PhastCons conservation (4way–100way), overlap with GC content and CGI (UCSC), and motif enrichment (TATA box: MA0108.3, INR: POL002.1, from JASPAR). Summits were extended ± 2 kb for conservation and GC content, and ± 50 bp for motif analysis; backgrounds were exon- and tCRE-masked. **f**, Distribution of CGI and TATA box around summits of major-strand and minor-strand tCREs of all the enhancers and enhancers divided into adjacent to promoter (enhancer-AP) and not adjacent (enhancer-NAP). **g**, Proportion of enhancer-like major-strand tCREs with antisense minor-strand tCREs and their regulatory features. **h**, Classification and naming schema for regulatory elements. **i**, Distance of different classes of enhancer with their closest promoter cCRE. **j**, Enrichment of chromatin states with CGI, CGInap, Mix, and TATA elements in enhancer-like tCREs across the 3 cell types. Enrichment test is performed by using Fisher's exact; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.